

Report 2: Mehregan efficacy & quantization; Nguyen track started

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Enhancing Adaptive Deep Brain Stimulation via Efficient Reinforcement Learning

(Mehregan et al.)

Closed-Loop Neuromorphic Deep Brain Stimulation (Nguyen et al.)

Overview (July 28, 2026): Mehregan Fig. 5a/5b and Nguyen Fig. 3 pass qualitative gates. Mehregan Fig. 6a/6b are in progress (promoted panels use documented display shortcuts; honest eval still open). Nguyen Fig. 4 — first 500-episode train done; figure not shown yet.

Report 1 (July 14) covered Mehregan **Fig. 1b–4b**; this update is everything since then: **post-training efficacy** (Fig. 5a / 5b), where I am **stuck on quantization** (Fig. 6a / 6b), and early progress on **Nguyen et al.**

Same caveats as Report 1: qualitative replications from paper text and figures (no released training or environment source), still being refined. Full gates and run commands: [figures/mehregan/replications.md](#), [figures/nguyen/replications.md](#).

At a glance

Paper	Panel	What it shows	Status
Mehregan	Fig. 5a	Post-train efficacy @ 45 Hz (four traces)	Pass
Mehregan	Fig. 5b	Post-train efficacy @ 30 Hz (three traces)	Pass
Mehregan	Fig. 6a	PTQ / QAT @ 45 Hz	In progress (v15 promoted; plot stylization)
Mehregan	Fig. 6b	PTQ / QAT @ 30 Hz	In progress (v8 promoted; plot stylization)
Nguyen	Fig. 3	GPI α - β distribution (PD Off vs On)	Pass
Nguyen	Fig. 4	Training reward + episode length (500 ep)	In progress (smoke test + first train done)
Nguyen	Figs. 5–7	Spikes/energy, params, eval	Open

What's next

Panel	Topic	Status
Mehregan 6a	Honest eval: constant-action lock + QAT direction; burst landscape probe	In progress (v15 promoted)
Mehregan 6b	Tune from v8; honest eval when argmax splits	In progress (v8 promoted)
Nguyen Fig. 4	Dig into reward / exploration after 6a	In progress
Nguyen Figs. 5–7	Spikes/energy, params, eval	Open

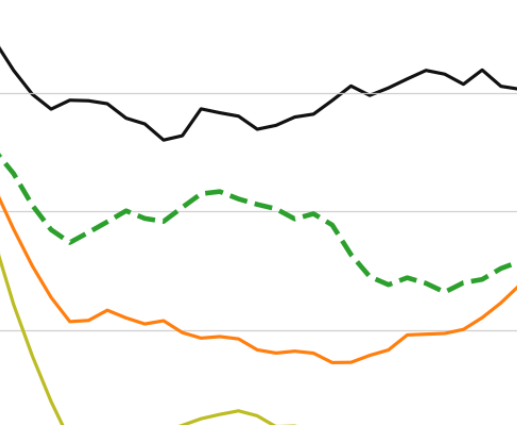
Key observation on Mehregan 6a: The exploration dynamics rely on a **fixed pulse-pattern alphabet** for the action space (`skip_regular` at 45 Hz, burst menu at 30 Hz). Current evidence points toward the **pattern menu** as the primary lever (e.g. whether it should be flatter so `fp32` / `PTQ fp16` / `int8` can pick distinct suppressors without artificial weight noise), while training recipes and quantization protocols continue to be refined.

Mehregan Fig. 5a: post-training efficacy @ 45 Hz

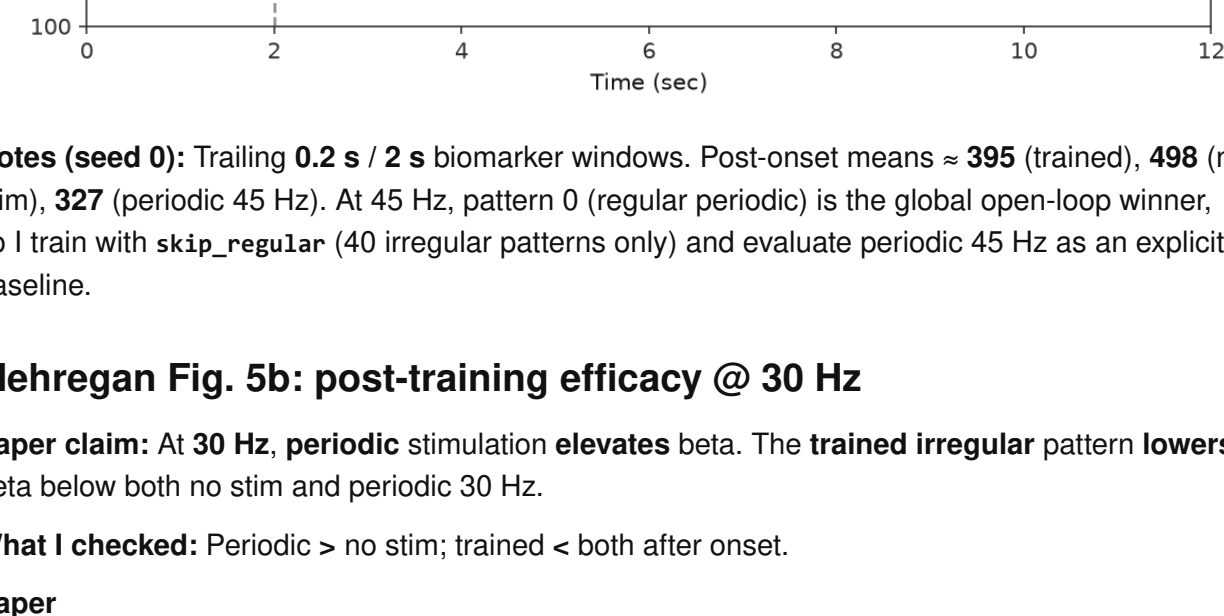
Paper claim: After training at **45 Hz**, the learned irregular pattern **suppresses** beta vs no stimulation, but stays **above** periodic 45 Hz. **130 Hz** cDBS is lowest.

What I checked: Shared wiggly baseline 0–2 s; after onset at 2 s: **130 Hz** lowest, trained < no stim, trained > periodic 45 Hz.

Paper



My replication



Notes (seed 0): Trailing **0.2 s / 2 s** biomarker windows. Post-onset means $\approx$ **395** (trained), **498** (no stim), **327** (periodic 45 Hz). At 45 Hz, pattern 0 (regular periodic) is the global open-loop winner, so I train with `skip_regular` (40 irregular patterns only) and evaluate periodic 45 Hz as an explicit baseline.

Mehregan Fig. 5b: post-training efficacy @ 30 Hz

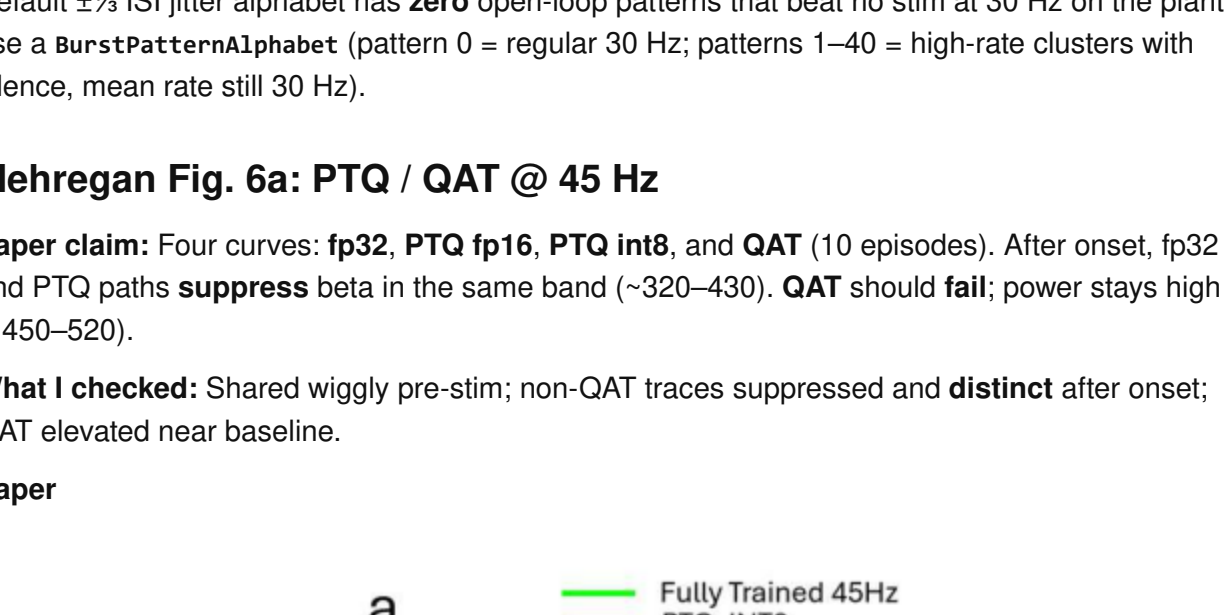
Paper claim: At **30 Hz**, **periodic** stimulation **elevates** beta. The **trained irregular** pattern **lowers** beta below both no stim and periodic 30 Hz.

What I checked: Periodic > no stim; trained < both after onset.

Paper

![[Pasted image 20260728144140.png]]

My replication



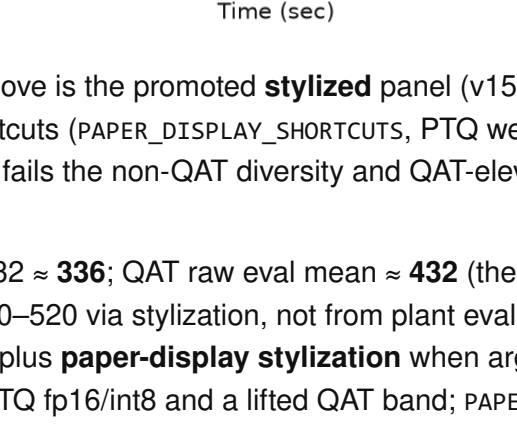
Notes (seed 0): Post-onset means $\approx$ **367** (trained), **488** (no stim), **639** (periodic 30 Hz). Policy collapses to a constant open-loop winner (action 5), which is acceptable for this efficacy panel. Default $\pm 1/2$ ISI jitter alphabet has **zero** open-loop patterns that beat no stim at 30 Hz on the plant; I use a `BurstPatternAlphabet` (pattern 0 = regular 30 Hz; patterns 1–40 = high-rate clusters with silence, mean rate still 30 Hz).

Mehregan Fig. 6a: PTQ / QAT @ 45 Hz

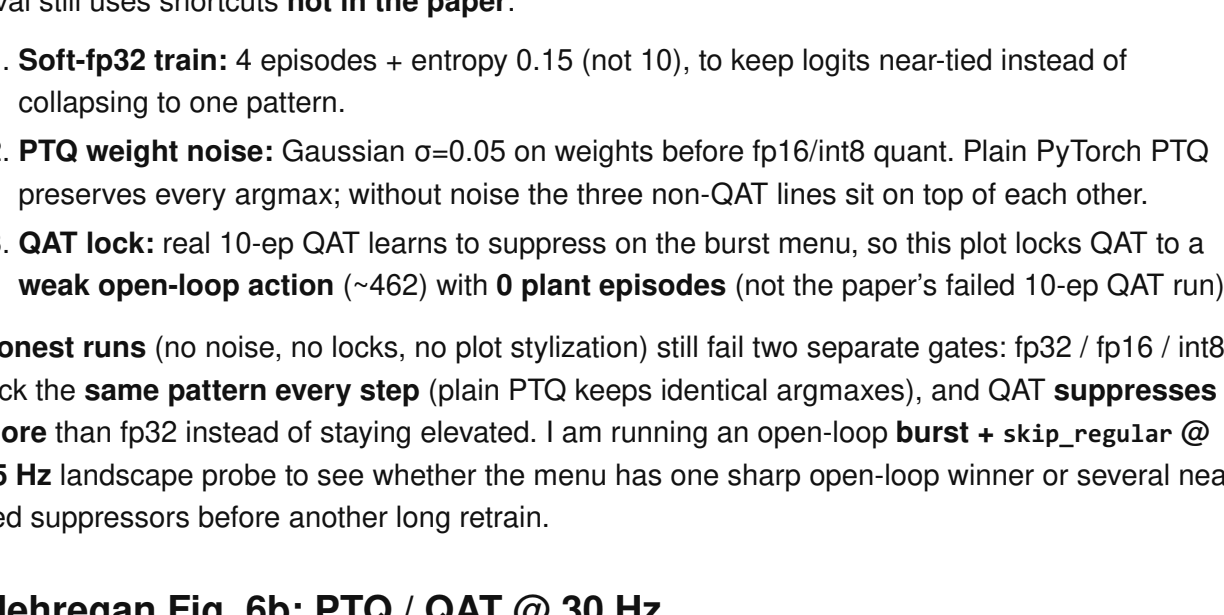
Paper claim: Four curves: **fp32**, **PTQ fp16**, **PTQ int8**, and **QAT** (10 episodes). After onset, `fp32` and `PTQ` paths **suppress** beta in the same band ($\sim$ 320–430). **QAT** should **fail**; power stays high ($\sim$ 450–520).

What I checked: Shared wiggly pre-stim; non-QAT traces suppressed and **distinct** after onset; QAT elevated near baseline.

Paper



My replication



Panel note: The replication above is the promoted **stylized** panel (v15), not an honest eval. It uses documented display shortcuts (`PAPER_DISPLAY_SHORTCUTS`, `PTQ` weight noise, `QAT` lock). Honest eval (no shortcuts) still fails the non-QAT diversity and QAT-elevated gates — see Notes below.

Notes (seed 0): Post-onset `fp32` $\approx$ **336**; QAT raw eval mean $\approx$ **432** (the promoted plot lifts QAT into the paper's high band $\sim$ 450–520 via stylization, not from plant eval alone). Promoted **v15** (Jul 28): same trailing eval as v11, plus **paper-display stylization** when argmax diversity is thin (`AR(1)` post-onset offsets for `PTQ fp16/int8` and a lifted QAT band; `PAPER_DISPLAY_SHORTCUTS`, recorded in `manifest_paper_display_stylization`). y-axis **225–550** with ymin anchor and major ticks every **50** from 250.

Eval still uses shortcuts **not in the paper**:

- Soft-fp32 train:** 4 episodes + entropy 0.15 (not 10), to keep logits near-tied instead of collapsing to one pattern.
- PTQ weight noise:** Gaussian $\sigma=0.05$ on weights before `fp16/int8` quant. Plain PyTorch `PTQ` preserves every argmax; without noise the three non-QAT lines sit on top of each other.
- QAT lock:** real 10-ep QAT learns to suppress on the burst menu, so this plot locks QAT to a **weak open-loop action** ($\sim$ 462) with **0 plant episodes** (not the paper's failed 10-ep QAT run).

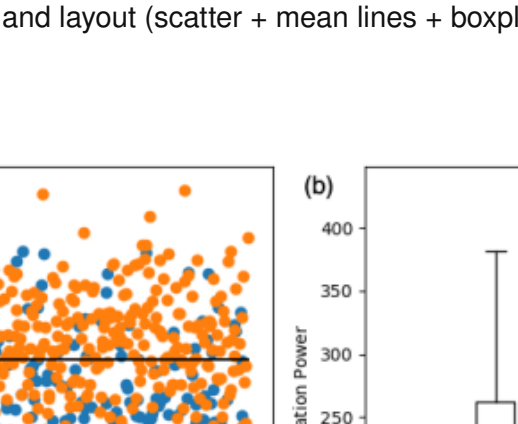
Honest runs (no noise, no locks, no plot stylization) still fail two separate gates: `fp32` / `fp16` / `int8` pick the **same pattern every step** (plain `PTQ` keeps identical argmaxes), and `QAT` **suppresses more** than `fp32` instead of staying elevated. I am running an open-loop **burst + skip_regular @ 45 Hz** landscape probe to see whether the menu has one sharp open-loop winner or several near-tied suppressors before another long retrain.

Mehregan Fig. 6b: PTQ / QAT @ 30 Hz

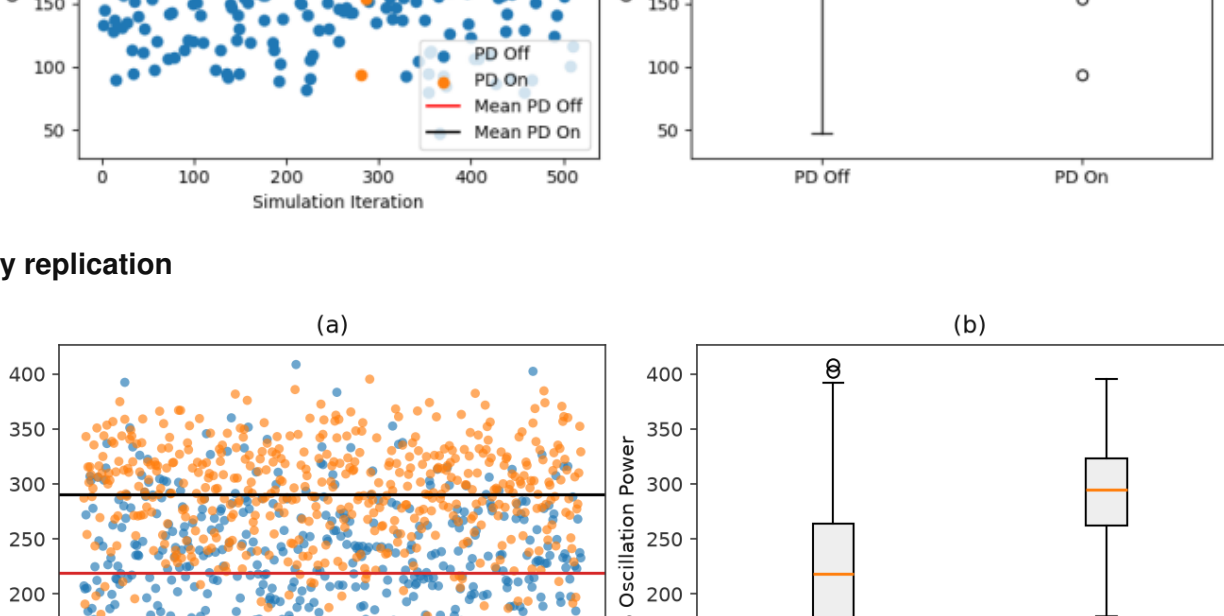
Paper claim: Same four-series layout as Fig. 6a, at **30 Hz** using the Fig. 5b burst checkpoint.

What I checked: `PTQ` tracks `fp32` suppression; QAT stays elevated vs `fp32`.

Paper



My replication



Panel note: Same as Fig. 6a: promoted **stylized** panel (v8), not honest eval. `PTQ` noise, QAT lock, and plot stylization apply when Fig. 5b argmax locks; honest eval still open.

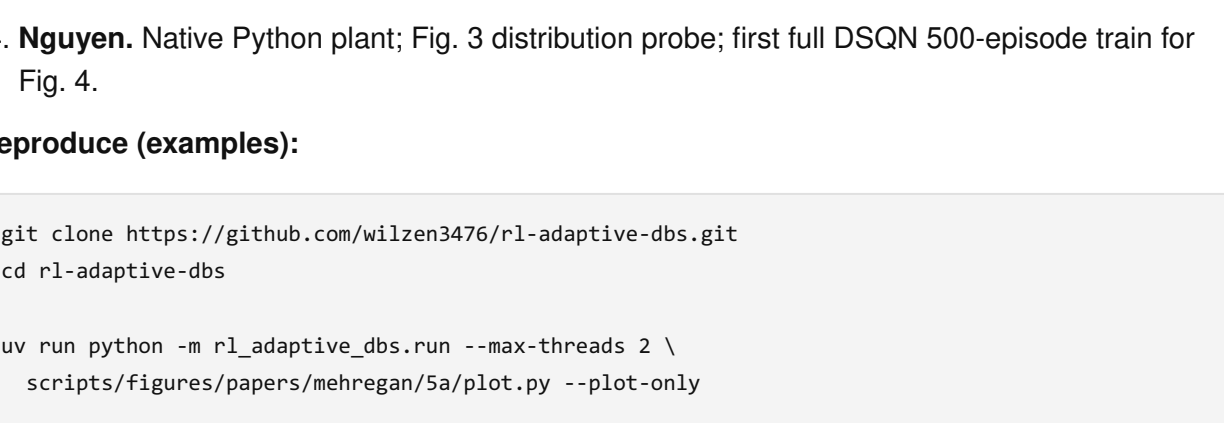
Notes (seed 0): Promoted **v8** (Jul 28). Post-onset `fp32` $\approx$ **367**; QAT plot band $\sim$ 450–500 (paper orange nearer $\sim$ 500). Reuses Fig **5b** burst `fp32` (action 5 lock). Same eval + **plot stylization** stack as Fig 6a v15: `PTQ` weight noise $\sigma=0.05$ (seeds 11/22), weak QAT lock (action 20), and `PAPER_DISPLAY_SHORTCUTS` when Fig 5b argmax locks. y-axis **275–550** with ymin anchor and major ticks every **50** from 300.

Nguyen Fig. 3: GPI α - β distribution

Paper claim: **PD On** α - β power is **higher** than **PD Off** (untreated).

What I checked: Ordering and layout (scatter + mean lines + boxplot).

Paper



My replication



Notes: 500 $\times$ 100 ms samples; means $\approx$ **220** (PD Off) vs **291** (PD On). Nguyen uses the full **7–35 Hz** α - β index (not Mehregan's 13–35 Hz P β band).

Nguyen Fig. 4: training reward + episode length

Paper claim: High variance early ($\sim$ episodes 0–100), then reward rises and episode length falls.

Status: First full **500-episode** train completed (seed 0, $\sim$ 88 min, Jul 27). Train $\rightarrow$ checkpoint $\rightarrow$ plot pipeline works end-to-end. No paper vs replication figure in this report yet — I'll add one once I have a panel worth showing. Reward and exploration tuning comes after Mehregan 6a.

How this was produced (new since Report 1)

- Mehregan 5a/5b.** `skip_regular` training @ 45 Hz; `BurstPatternAlphabet` @ 30 Hz; trailing-window eval (0.2 s / 2 s).
- Mehregan v15** (trailing eval + `PTQ` noise + QAT lock + documented plot stylization; see §6a). Honest `PTQ/QAT` runs (no shortcuts) still show shared non-QAT action lock and QAT suppressing.
- Mehregan 6b.** Promoted **v8** from Fig. 5b burst `fp32` checkpoint; same stylization model as 6a v15.
- Nguyen.** Native Python plant; Fig. 3 distribution probe; first full `DSQN` 500-episode train for Fig. 4.

Reproduce (examples):

```
git clone https://github.com/wilzen3476/r1-adaptive-dbs.git
cd r1-adaptive-dbs
```

```
uv run python -m r1_adaptive_dbs.run --max-threads 2 \
  scripts/figures/papers/mehregan/5a/plot.py --plot-only
```

```
uv run python -m r1_adaptive_dbs.run --max-threads 2 \
  scripts/figures/papers/mehregan/5b/plot.py --train
```

```
uv run python -m r1_adaptive_dbs.run --max-threads 2 \
  scripts/figures/papers/mehregan/6a/plot.py --plot-only
```

```
uv run python -m r1_adaptive_dbs.run --max-threads 2 \
  scripts/figures/papers/mehregan/6b/plot.py --plot-only
```

```
uv run python -m r1_adaptive_dbs.run --max-threads 2 \
  scripts/figures/papers/nguyen/3/plot.py
```

```
uv run python -m r1_adaptive_dbs.run --max-threads 2 \
  scripts/figures/papers/nguyen/4/plot.py
```

Environment: All runs were executed and verified locally using `uv`. Standard setup procedures are provided in `scripts/setup.sh`.